Supporting Information

Photophysical properties, halochromism and applications of novel Lophines as inkless writing material and in pH responsive fluorescent calcium alginate hydrogel beads

Subhash B. Patel,[†‡] Vidhi V. Bhopekar,[‡] Bharvi M. Patel,[‡] Zhe Wang,[±] Tadashi Mori,[±] Gajendrasinh B. Raolji,[†] Sasikumar T. Kuttappan,[†] Paresh D. Zalavadiya,[†] Sachin G. Modha[‡]*

†Piramal Pharma Limited, Plot No. 18 Pharmaceutical Special Economic Zone, Village Matoda, Taluka: Sanand, 382213, Ahmedabad, Gujarat, India

‡Laboratory of Organic Chemistry, Tarsadia Institute of Chemical Science, Uka Tarsadia University, Maliba Campus, Gopal Vidyanagar, Bardoli-Mahuva Road, Tarsadi 394650, Surat, Gujarat, India.

[±]Division of Applied Chemistry, Graduate School of Engineering, The University of Osaka, 2-4 Yamada-Oka, Suita, Osaka 565-0871, Japan.

Corresponding Author

***Email:** sachinmodha@gmail.com

Characterization of compounds **3, 4** and **5** S2

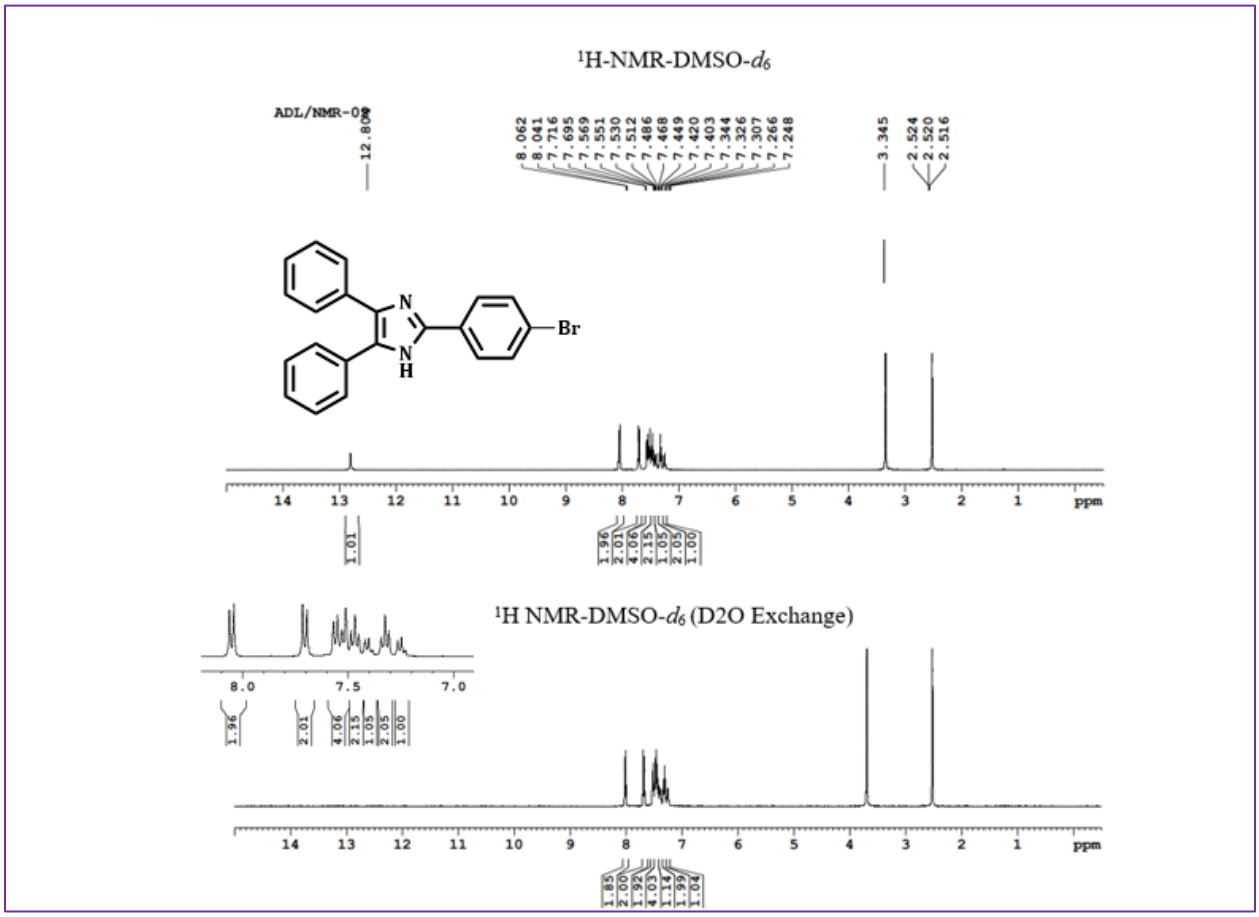
Absorption and emission spectra in DMSO solvent.....S15

Absorption and emission spectra for solvatochromism and halochromism study S16

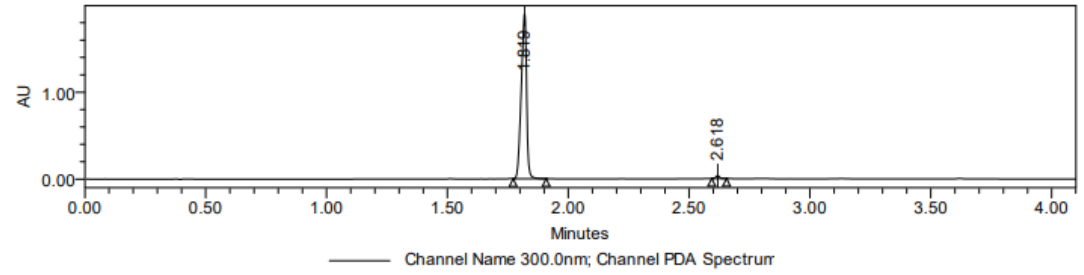
Theoretical calculations S18

Stability study under alkaline and acidic pH S25

^{1}H-NMR (400 MHz) Spectra of compound 2-(4-bromophenyl)-4,5-diphenyl-1*H*-imidazole in DMSO-d_6 (**3**)



LCMS Spectra of Compound 2-(4-bromophenyl)-4,5-diphenyl-1*H*-imidazole (**3**)

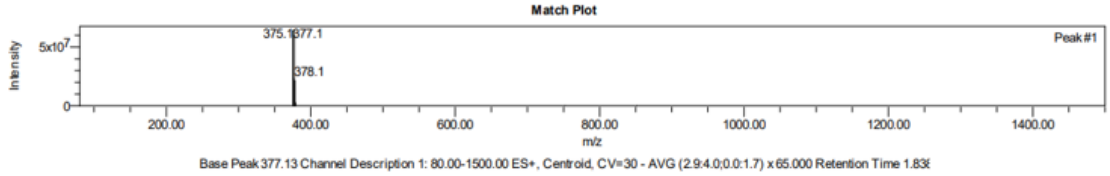


Peak Results
Channel: PDA Spectrum

	Retention Time (min)	Base Peak (m/z)	Height (µV)	Area (µV*sec)	% Area	Channel	Channel Name
1	1.818		950819	1749199	93.82	PDA Spectrum	210.0nm
2	1.819		1905573	2927562	98.82	PDA Spectrum	300.0nm

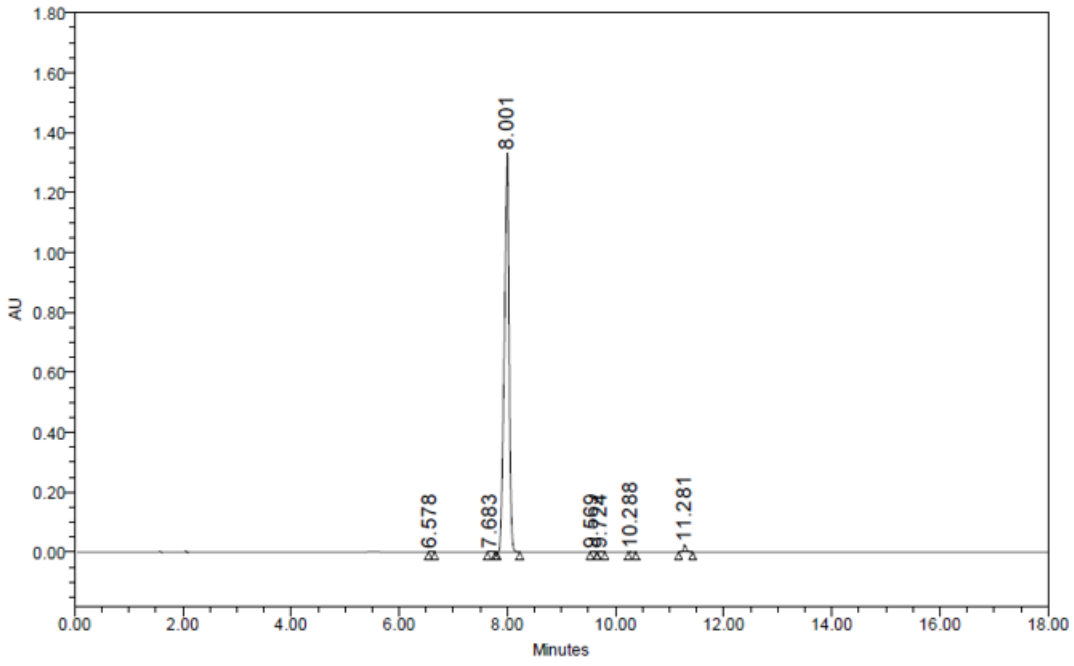
Peak Results
Channel: SQ 1: MS Scan

	Retention Time (min)	Base Peak (m/z)	Height (µV)	Area (µV*sec)	% Area	Channel	Channel Name
1	1.838	377.13	244585122	963242352	66.14	SQ 1: MS Scan	MS TIC



Base Peak 377.13 Channel Description 1: 80.00-1500.00 ES+, Centroid, CV=30 - AVG (2.9:4.0;0.0:1.7) x 65.000 Retention Time 1.83

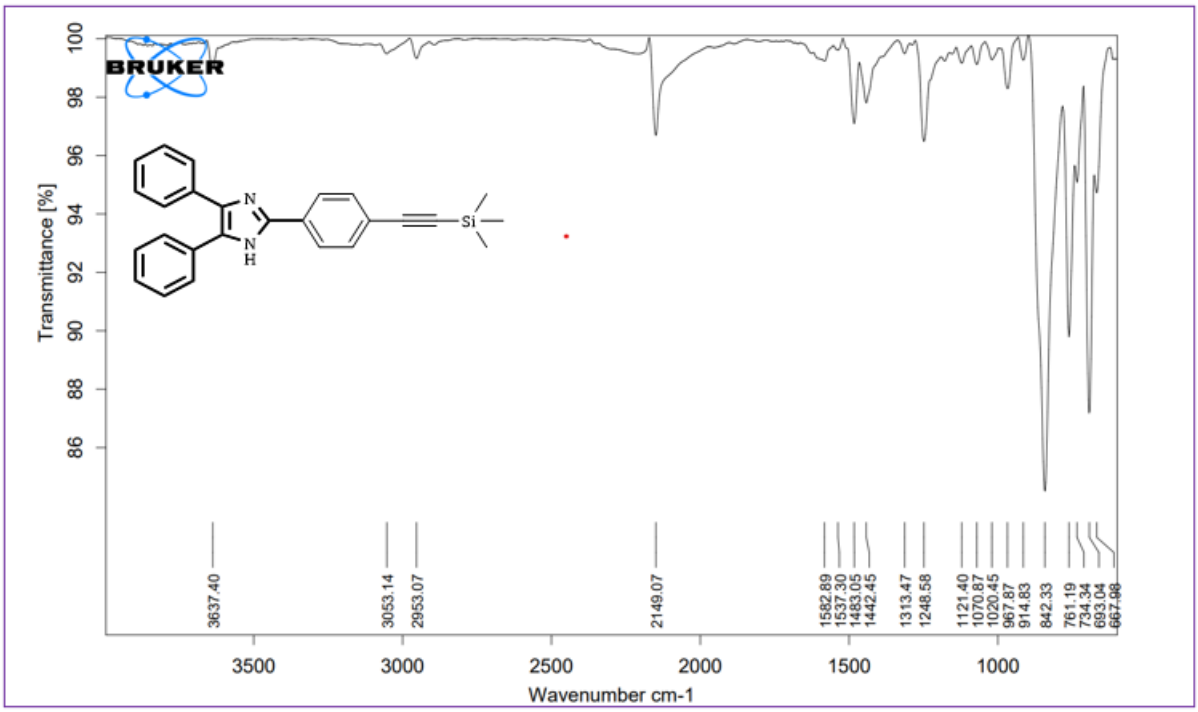
HPLC Spectra of Compound 2-(4-bromophenyl)-4,5-diphenyl-1*H*-imidazole (**3**)



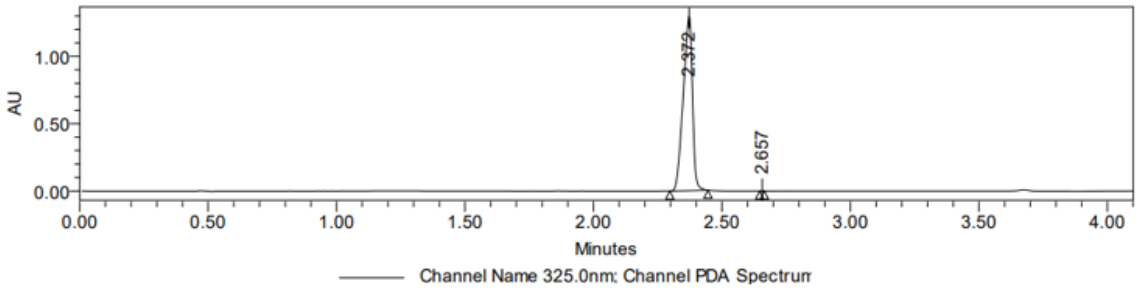
Peak Results

	RT	Area (μV*sec)	Height (μV)	% Area
1	6.578	7128	2636	0.09
2	7.683	11828	3415	0.14
3	8.001	8159948	1330914	98.59
4	9.569	1983	569	0.02
5	9.724	2239	557	0.03
6	10.288	1692	476	0.02
7	11.281	91623	22728	1.11

IR Spectra of Compound 4,5-diphenyl-2-(4-((trimethylsilyl)ethynyl)phenyl)-1*H*-imidazole (**4**)



LCMS Spectra of Compound 4,5-diphenyl-2-(4-((trimethylsilyl)ethynyl)phenyl)-1*H*-imidazole (**4**)

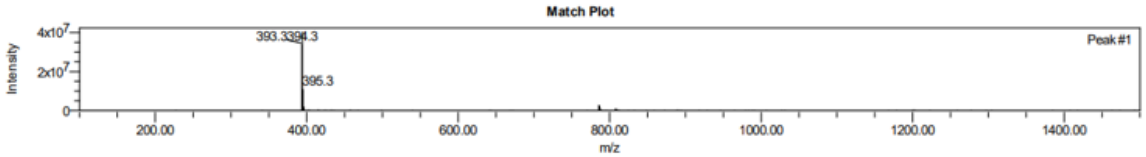


Channel: PDA Spectrum

	Retention Time (min)	Base Peak (m/z)	Height (µV)	Area (µV*sec)	% Area	Channel	Channel Name
1	2.214		7373	9543	0.52	PDA Spectrum	220.0nm
2	2.372		1299286	3228076	99.99	PDA Spectrum	325.0nm
3	2.372		695828	1815298	99.48	PDA Spectrum	220.0nm
4	2.657		474	336	0.01	PDA Spectrum	325.0nm

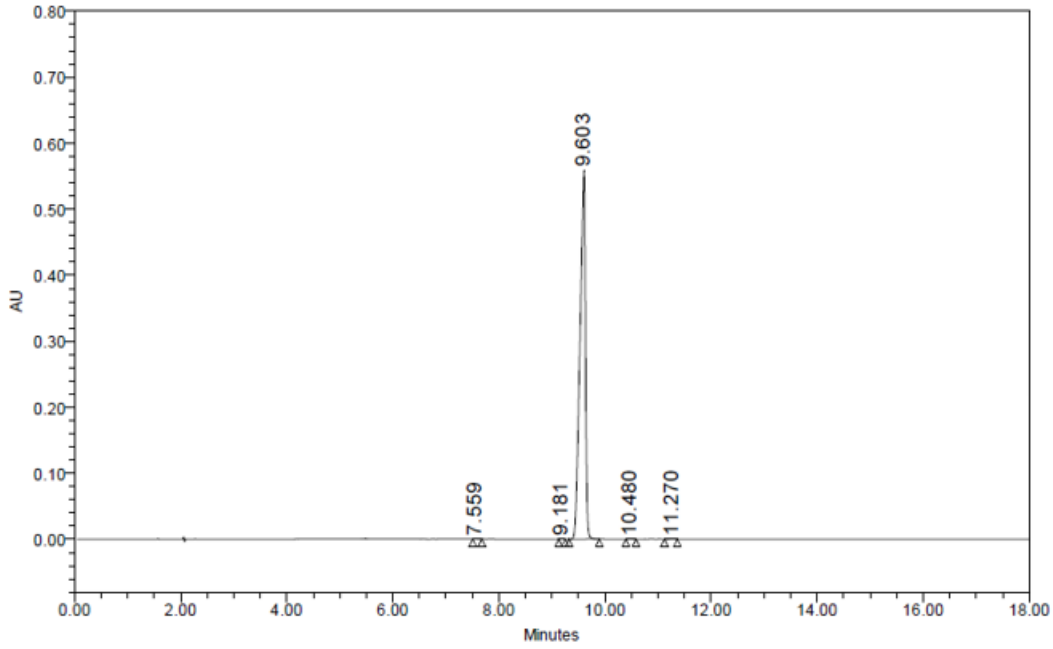
Peak Results
Channel: SQ 1: MS Scan

	Retention Time (min)	Base Peak (m/z)	Height (µV)	Area (µV*sec)	% Area	Channel	Channel Name
1	2.395	394.29	109060777	565965437	99.97	SQ 1: MS Scan	MS TIC
2	2.524	130.01	169758	158300	0.03	SQ 1: MS Scan	MS TIC



Base Peak 394.29 Channel Description 1: 80.00-1500.00 ES+, Centroid, CV=30 - AVG (2.6:3.7;0.1:2.3) x 99.000 Retention Time 2.39

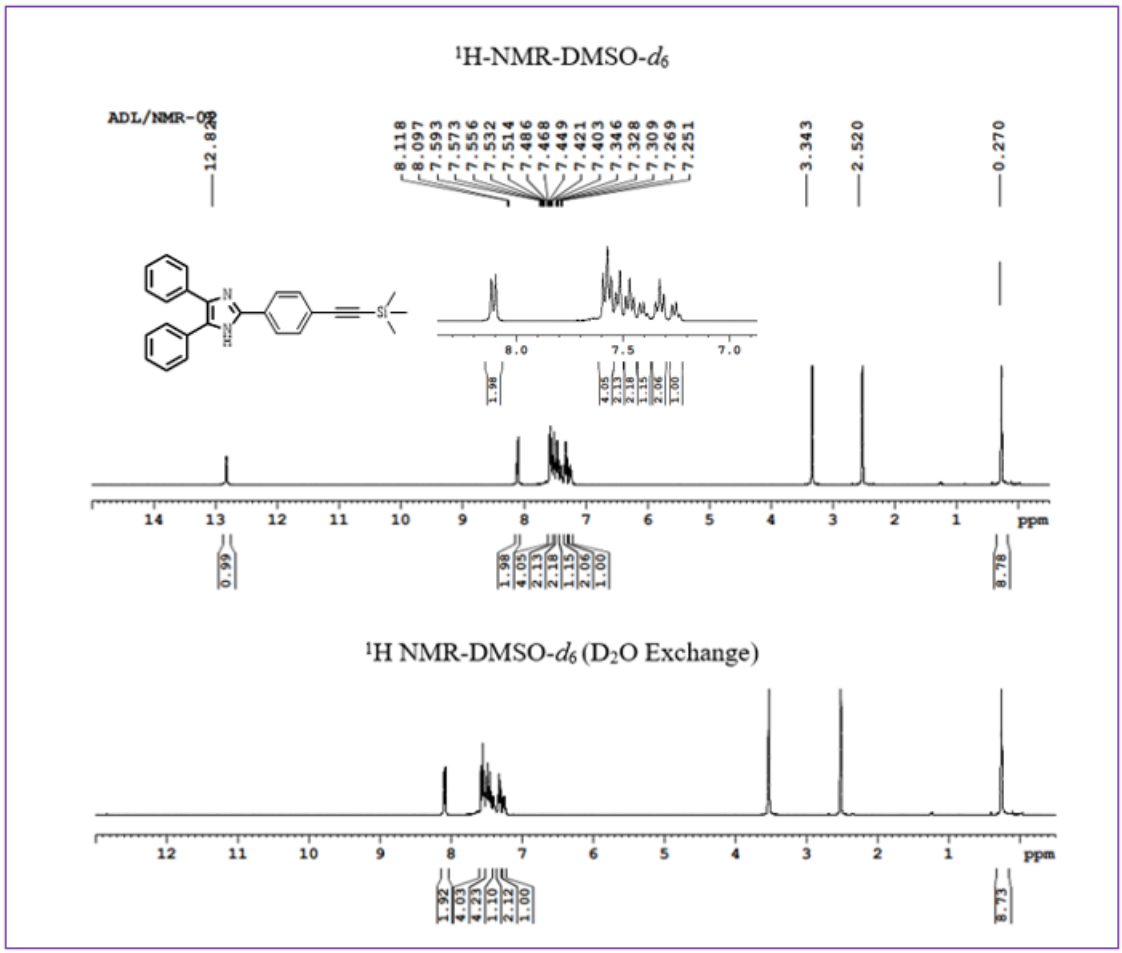
HPLC Spectra of Compound 4,5-diphenyl-2-(4-((trimethylsilyl)ethynyl)phenyl)-1*H*-imidazole (**4**)



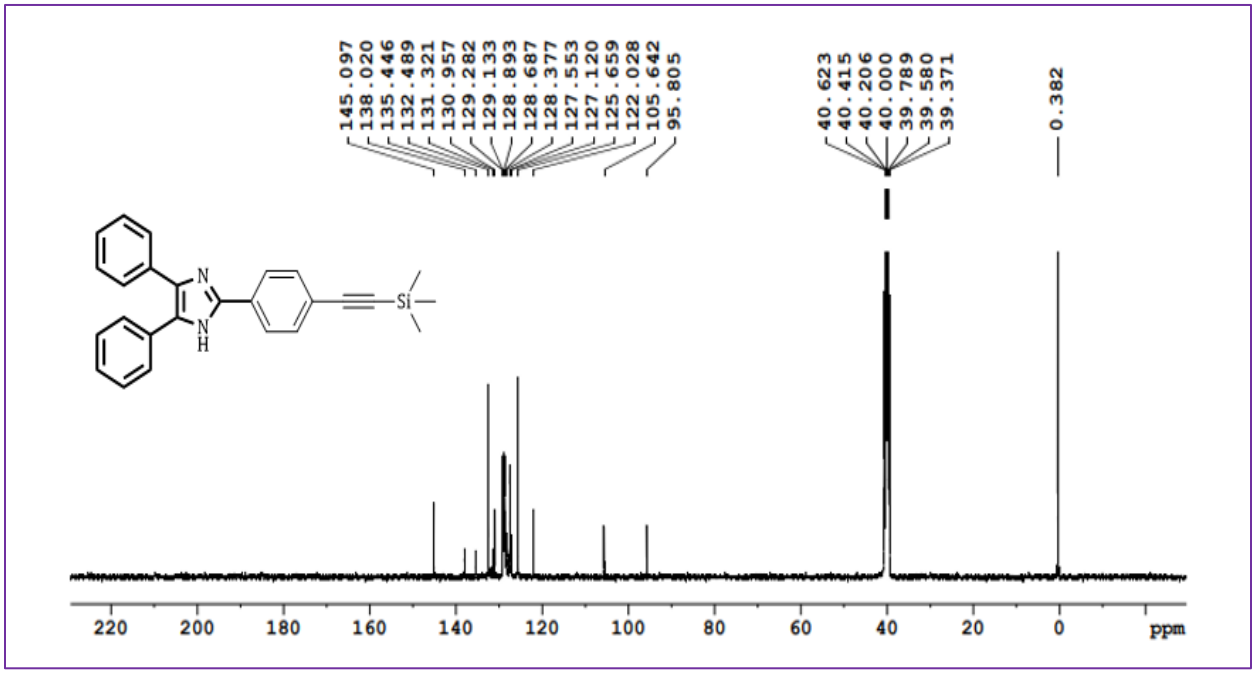
Peak Results

	RT	Area (µV*sec)	Height (µV)	% Area
1	7.559	2710	800	0.07
2	9.181	2322	745	0.06
3	9.603	3945680	558627	99.76
4	10.480	1483	386	0.04
5	11.270	2831	528	0.07

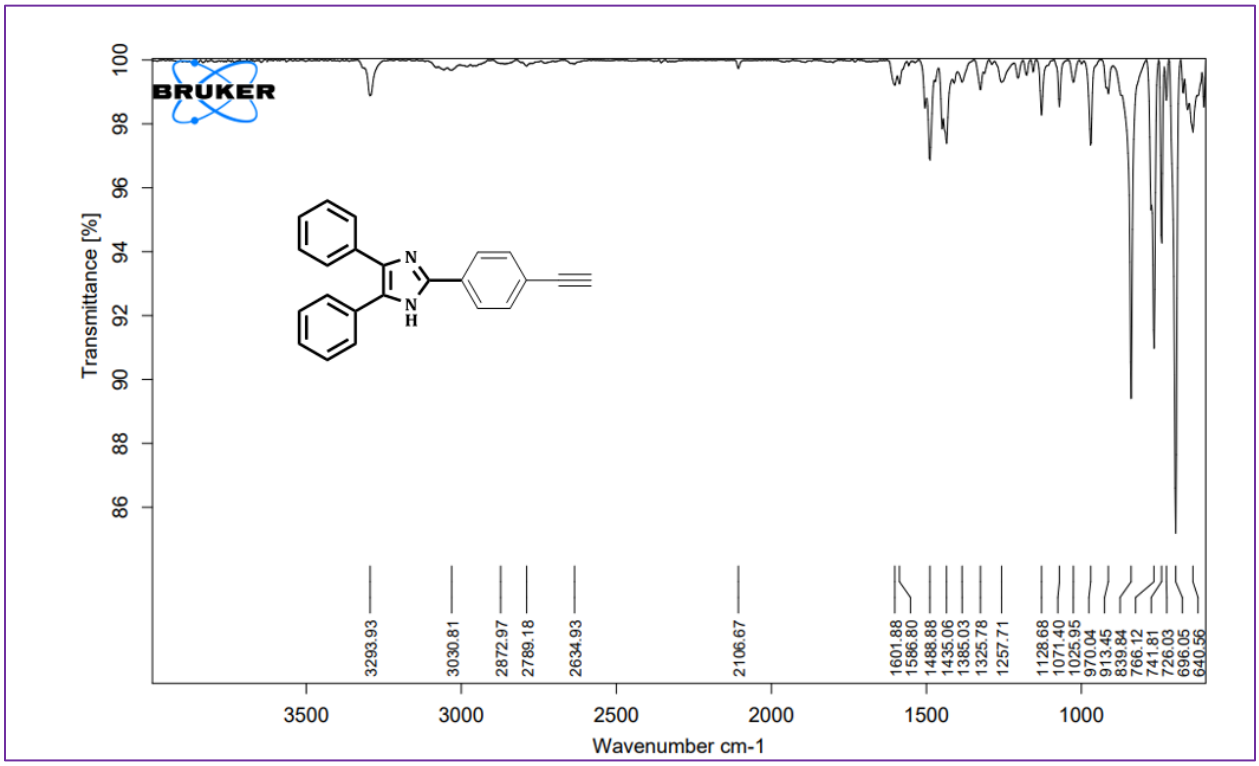
^{1}H-NMR Spectra of Compound 4,5-diphenyl-2-(4-((trimethylsilyl)ethynyl)phenyl)-1*H*-imidazole (**4**)



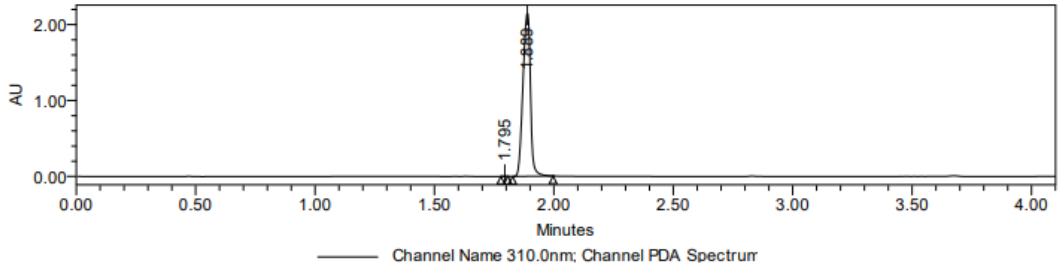
^{13}C-NMR spectra of Compound 4,5-diphenyl-2-(4-((trimethylsilyl)ethynyl)phenyl)-1*H*-imidazole (4)



IR Spectra of Compound 2-(4-ethynylphenyl)-4,5-diphenyl-1*H*-imidazole (**5**)



LCMS spectra of Compound 2-(4-ethynylphenyl)-4,5-diphenyl-1*H*-imidazole (**5**)

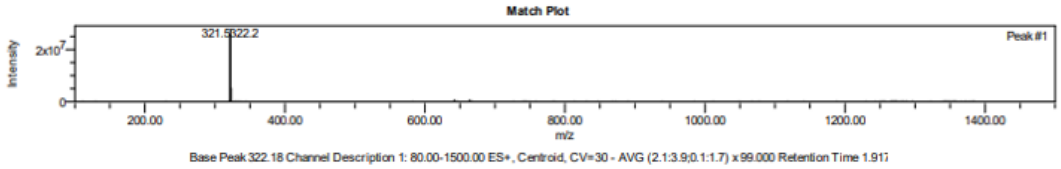


Channel: PDA Spectrum

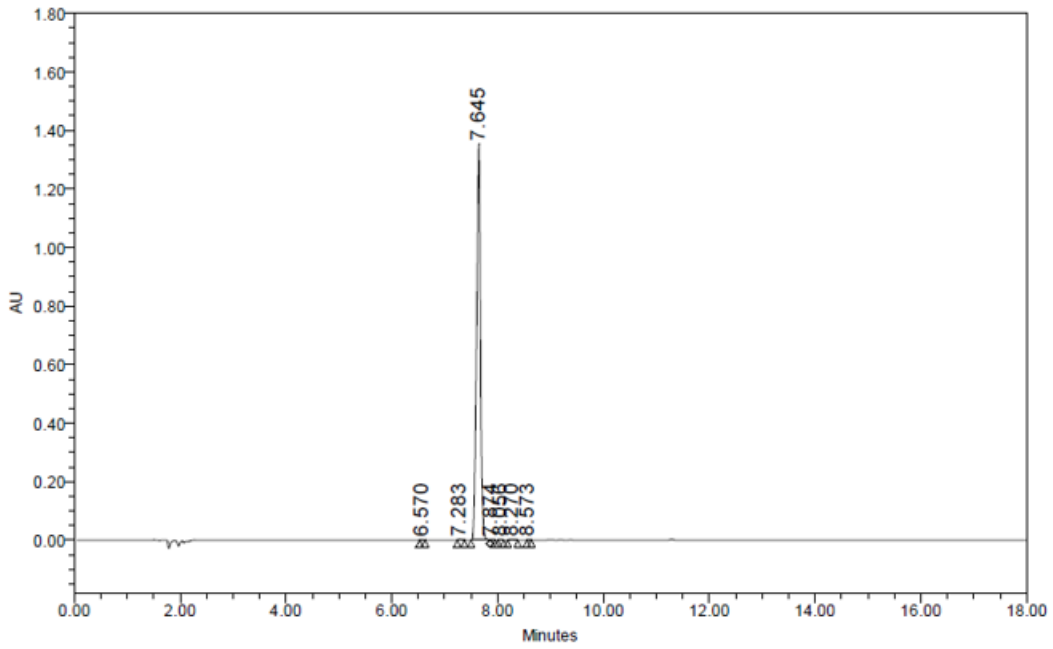
	Retention Time (min)	Base Peak (m/z)	Height (µV)	Area (µV*sec)	% Area	Channel	Channel Name
1	1.795		3300	3010	0.06	PDA Spectrum	310.0nm
2	1.888		1097951	2613310	99.48	PDA Spectrum	220.0nm
3	1.889		2148197	4685008	99.94	PDA Spectrum	310.0nm
4	2.218		10503	13725	0.52	PDA Spectrum	220.0nm

Peak Results
Channel: SQ 1: MS Scan

	Retention Time (min)	Base Peak (m/z)	Height (µV)	Area (µV*sec)	% Area	Channel	Channel Name
1	1.917	322.18	92214891	284099346	100.00	SQ 1: MS Scan	MS TIC



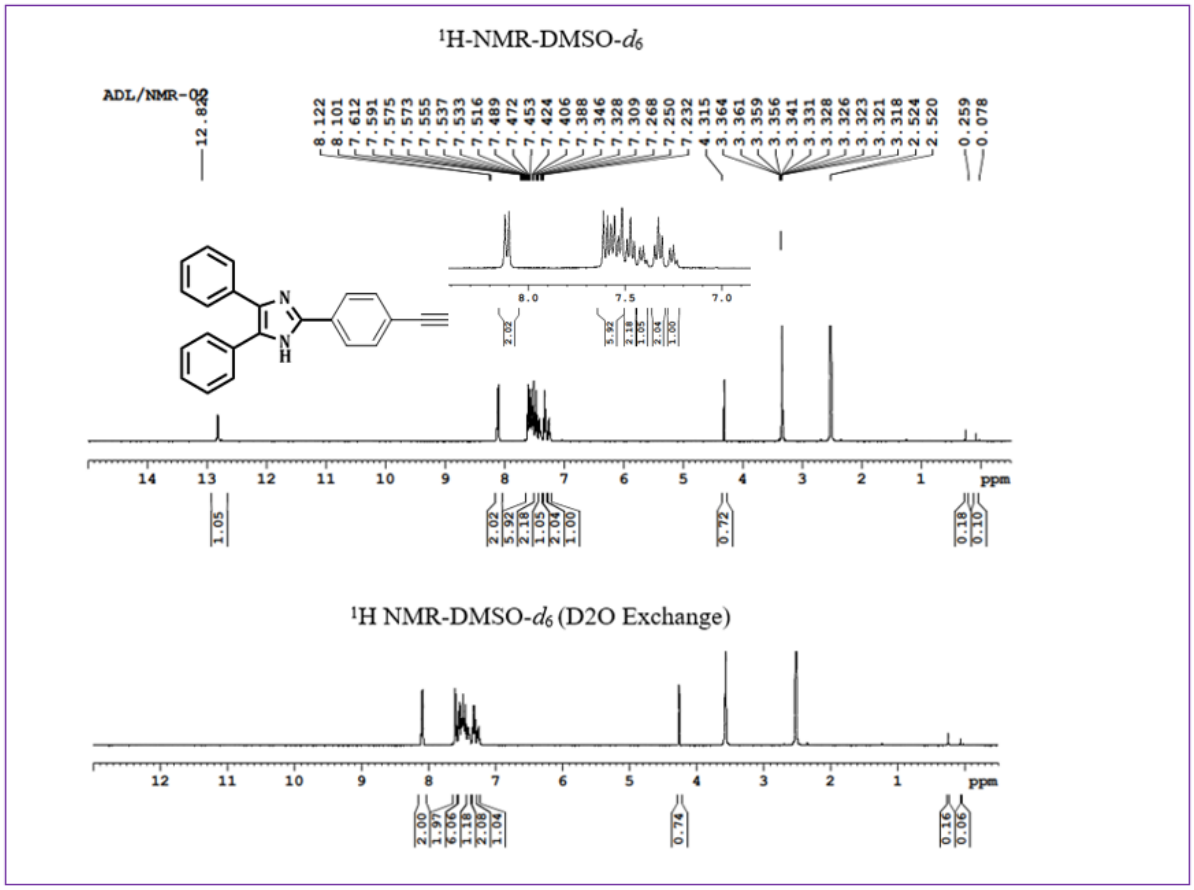
HPLC Spectra of Compound 2-(4-ethynylphenyl)-4,5-diphenyl-1*H*-imidazole (**5**)



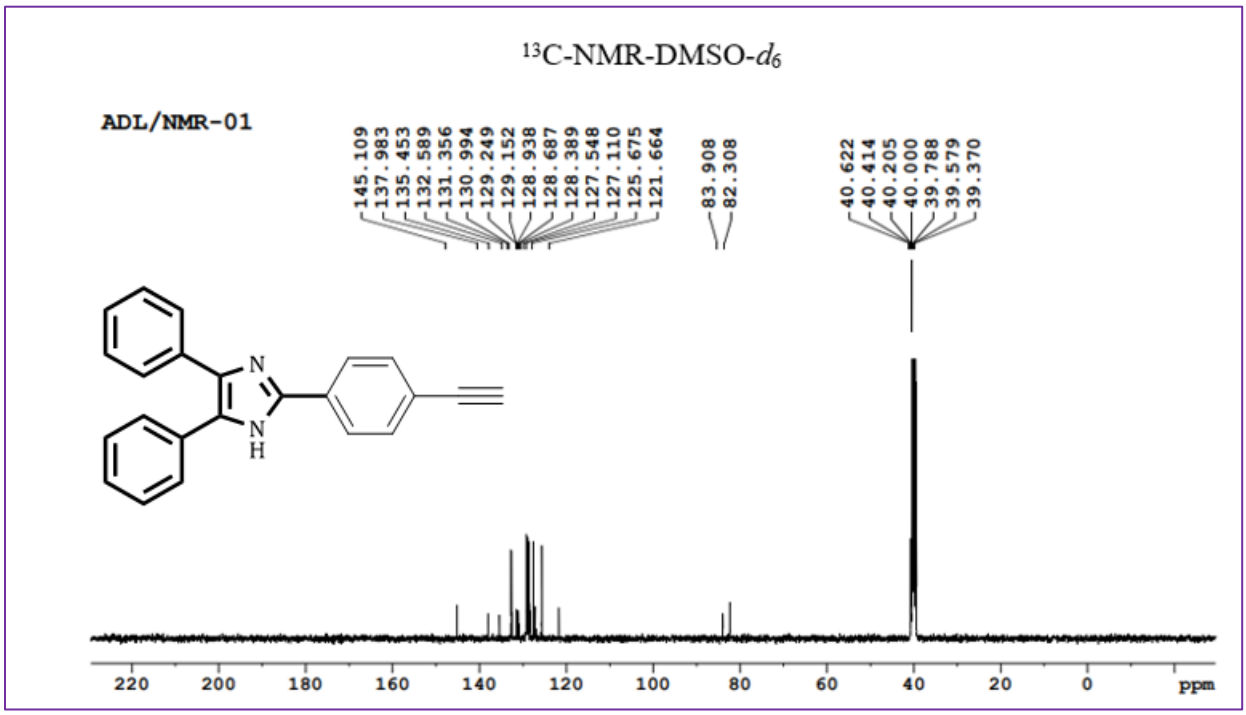
Peak Results

	RT	Area (μV*sec)	Height (μV)	% Area
1	6.570	3710	1414	0.05
2	7.283	7845	2491	0.11
3	7.645	6965000	1354528	99.72
4	7.874	2541	761	0.04
5	8.056	1514	488	0.02
6	8.270	3907	1249	0.06
7	8.573	371	163	0.01

^{1}H-NMR Spectra of Compound 2-(4-ethynylphenyl)-4,5-diphenyl-1*H*-imidazole (**5**)



^{13}C-NMR spectra of compound 2-(4-ethynylphenyl)-4,5-diphenyl-1*H*-imidazole (**5**)



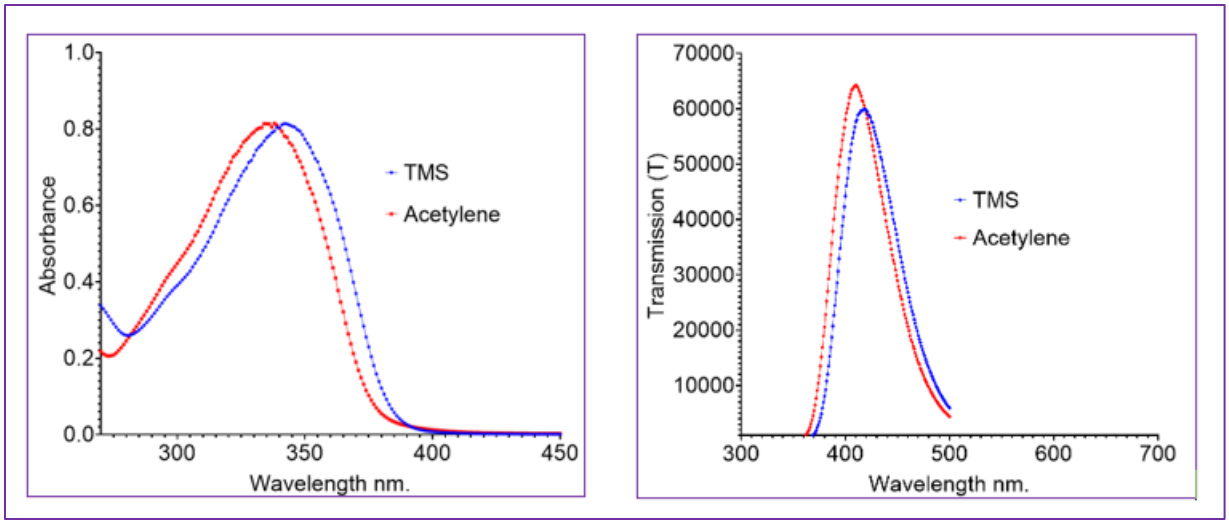


Fig. S1. Absorption spectra and non-normalized fluorescence emission spectra of compounds **4** and **5** measured in DMSO solvent.

Table S1. The data of absorption, emission for compound 4,5-diphenyl-2-(4-((trimethylsilyl)ethynyl)phenyl)-1*H*-imidazole (**4**) in different solvents.

No.	Solvent	Dielectric constant	λ_{abs} (nm)	λ_{em} (nm)	ε ($M^{-1}cm^{-1}$)	E_{gap} (eV)	Stokes shift (cm^{-1})
1	MeOH	32.70	332	432	46.7 x 10^3	3.26	6091
2	IPA	19.92	336	427	40.9 x 10^3	3.23	5650
3	DMSO	46.68	342	422	40.7 x 10^3	3.21	5205
4	DMF	36.71	343	439	38.4 x 10^3	3.16	5711
5	ACN	37.5	335	433	45.6 x 10^3	3.18	5641
6	EtOAc	6.02	339	432	42.4 x 10^3	3.23	6006
7	DCM	8.93	338	410	34.5 x 10^3	3.22	4679
8	Dioxane	2.25	340	424	45.1 x 10^3	3.19	5151

λ_{abs} = Maximum absorption wavelength, λ_{em} = Maximum emission wavelength, ε= Molar absorption coefficient, E_{gap} = Optical band gap.

Table S2. The data of absorption, emission for compound 2-(4-ethynylphenyl)-4,5-diphenyl-1*H*-imidazole (**5**) in different solvents.

No.	Solvent	Dielectric constant	λ_{abs} (nm)	λ_{em} (nm)	ε ($M^{-1}cm^{-1}$)	E_{gap} (eV)	Stokes shift (cm^{-1})
1	MeOH	32.70	322	436	52.5 x 10^3	3.12	4536
2	IPA	19.92	330	432	47.1 x 10^3	3.10	4249
3	DMSO	46.68	338	427	40.7 x 10^3	3.10	3978
4	DMF	36.71	338	445	50.1 x 10^3	3.07	3983
5	ACN	37.5	332	436	52.4 x 10^3	3.08	3659
6	EtOAc	6.02	332	434	48.7 x 10^3	3.14	3840
7	DCM	8.93	334	428	48.4 x 10^3	3.13	3302
8	Dioxane	2.25	335	427	47.8 x 10^3	3.09	3106

λ_{abs} = Maximum absorption wavelength, λ_{em} = Maximum emission wavelength, ε= Molar absorption coefficient, E_{gap} = Optical band gap.

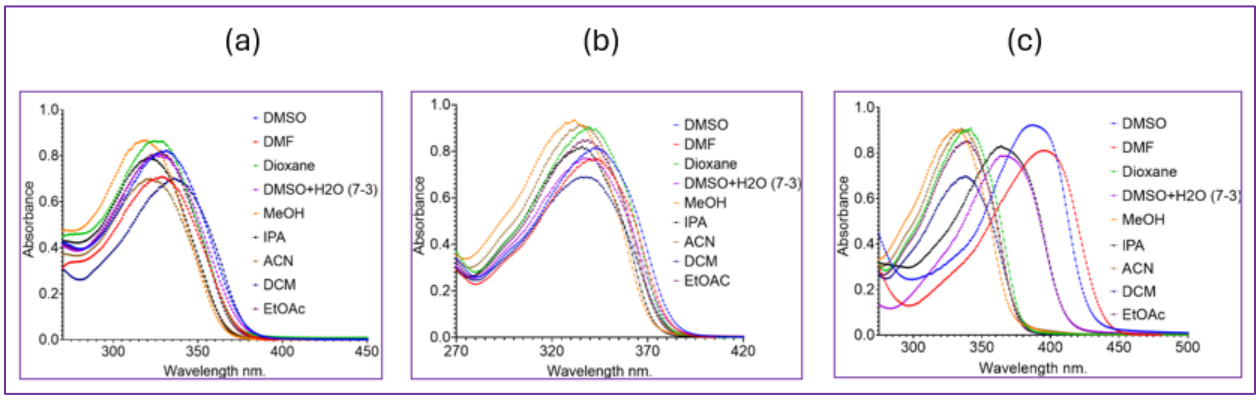


Fig. S2. Absorption spectra of **4** in various solvents: (a) acidic pH, (b) neutral pH and (c) basic pH.

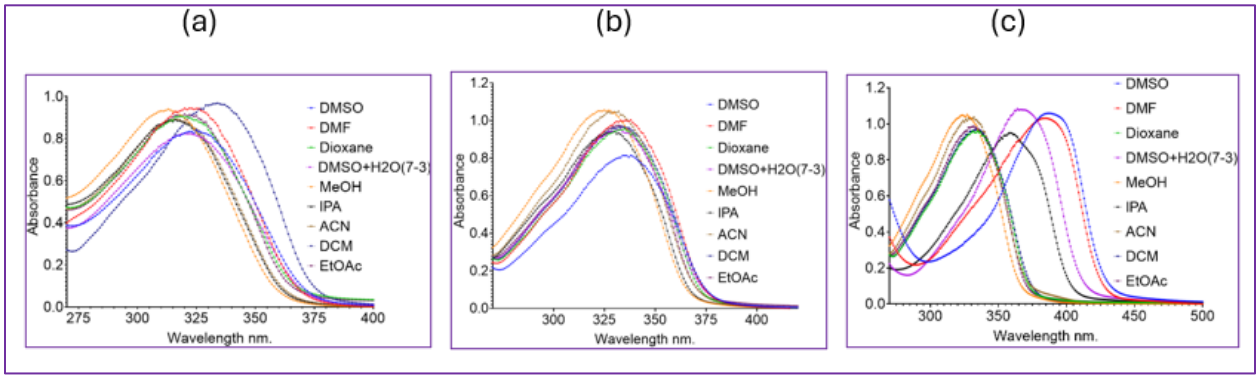


Fig. S3. Absorption spectra of **5** in various solvents: (a) acidic pH, (b) neutral pH and (c) basic pH.

Table S3. Details of $S_0 \rightarrow S_1$ and $S_1 \rightarrow S_0$ transition calculated at ωB97X-D/def2-TZVP level of theory.

	UV: $S_0 \rightarrow S_1$	**FL: $S_1 \rightarrow S_0$**
4 (neutral)	303.14 nm (4.0899 eV), f = 1.2413 HOMO → LUMO (86.41%) HOMO–1 → LUMO (3.91%)	376.66 nm (3.2917 eV), f = 1.1569 LUMO → HOMO (92.78%)
4 (anion)	354.50 nm (3.4975 eV), f = 1.3486 HOMO → LUMO (89.13%) HOMO–3 → LUMO (2.46%)	395.74 nm (3.1329 eV), f = 1.5287 LUMO → HOMO (92.25%)
5 (neutral)	298.71 nm (4.1507 eV), f = 0.9779 HOMO → LUMO (86.94%) HOMO–1 → LUMO (3.08%) HOMO → LUMO+1 (2.78%)	374.85 nm (3.3075 eV), f = 0.8786 LUMO → HOMO (92.66%) LUMO → HOMO–1 (2.04%)
5 (anion)	340.84 nm (3.6376 eV), f = 1.1151 HOMO → LUMO (91.65%) HOMO → LUMO+11 (2.24%)	378.18 nm (3.2784 eV), f = 1.2551 LUMO → HOMO (94.04%)

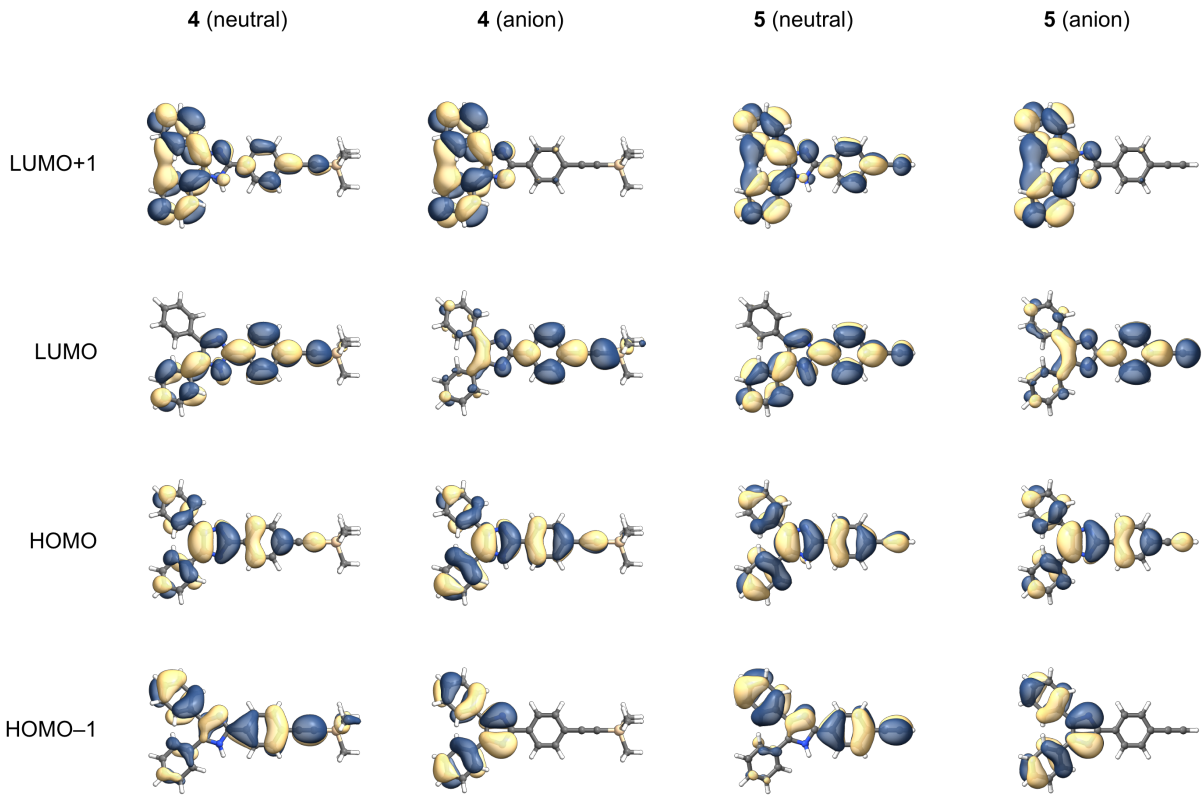


Fig. S4 Molecular orbitals calculated at the ωB97X-D/def2-TZVP level of theory, isovalue = 0.02.

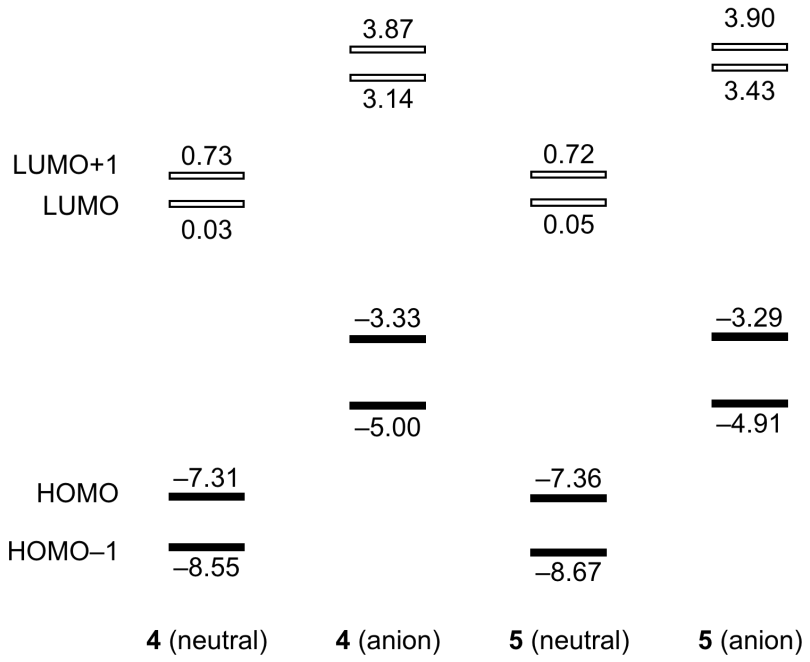


Fig. S5 Molecular orbital energy levels (in eV) calculated at the ωB97X-D/def2-TZVP level of theory.

Coordinates of Computed Geometries

4 (neutral, S_0)

Si	-7.86367800	-0.00408800	-0.02293300
N	1.60600300	1.10544000	0.06868200
N	1.66395100	-1.08506900	0.01167200
C	0.87374100	0.01923000	0.03942300
C	2.97436200	-0.67587300	0.00177000
C	2.91293600	0.69952400	0.05504400
C	4.00540300	1.68090200	0.11750600
C	6.04803300	3.58290100	0.21673800
C	3.84497300	2.92662400	-0.48580700
C	5.19697500	1.40866700	0.78674700
C	6.21149800	2.35048500	0.83139800
C	4.85874200	3.86871200	-0.43755500
C	4.07142800	-1.64220100	-0.09403600
C	6.14083700	-3.50541900	-0.30352400
C	5.15230900	-1.40920500	-0.94388000
C	4.04280300	-2.82368500	0.64624500
C	5.06774200	-3.75029700	0.53850800
C	6.17988400	-2.33049100	-1.04145900
C	-0.58790800	-0.01718200	0.03873900
C	-3.38957900	-0.01360700	0.01810300
C	-1.30180700	-1.20326500	0.19779500
C	-1.29458100	1.17398800	-0.12344800
C	-2.67376900	1.17705900	-0.13314900
C	-2.68337500	-1.20488400	0.18547500
C	-4.81693300	-0.01204800	0.00544500
C	-6.02555300	-0.01045100	-0.00574700
C	-8.43654500	0.92006600	-1.54325300
C	-8.46730300	0.84833400	1.52711600
C	-8.45704700	-1.77609600	-0.06973900
H	1.36369100	-2.03435600	-0.11702900
H	2.91268800	3.14726300	-0.98901900
H	5.32678000	0.45664100	1.28517900
H	7.12998500	2.12389500	1.35828200
H	4.71967400	4.83136300	-0.91360200
H	6.84043000	4.31973800	0.25437300
H	5.18015100	-0.49872300	-1.52813500
H	3.22415200	-3.00735700	1.33247300
H	5.03237500	-4.66057300	1.12355100
H	7.01268100	-2.13422400	-1.70472300
H	6.94434900	-4.22613600	-0.38468000
H	-0.78706000	-2.14412300	0.35124200
H	-0.73892000	2.09457500	-0.23950700
H	-3.21424100	2.10568600	-0.26011800
H	-3.22802300	-2.13109400	0.31146200
H	-9.52796200	0.94552700	-1.58987600
H	-8.06932900	0.44198200	-2.45323900
H	-8.07413800	1.94971500	-1.53440200
H	-9.55945900	0.87266000	1.55267400
H	-8.10450300	1.87678500	1.57413200
H	-8.11873000	0.32770600	2.42094600
H	-9.54890700	-1.81639500	-0.08157800
H	-8.10770800	-2.32855700	0.80457600
H	-8.08976100	-2.28809000	-0.96112800

4 (anion, S_0)

Si	-7.80568900	-0.00037100	-0.00004300
N	1.64037000	1.13315100	-0.00799000
N	1.63928300	-1.13163300	0.00814100
C	0.92046300	0.00110200	0.00010600
C	2.92011100	-0.70396800	-0.00203800
C	2.92078600	0.70423700	0.00202300
C	4.01768400	1.67610600	0.06987000
C	6.06404000	3.60166900	0.20475700
C	3.87125500	2.93129100	-0.52876600
C	5.20776500	1.42012400	0.75588000
C	6.21878600	2.36560300	0.81608100
C	4.87838500	3.87907000	-0.46275400
C	4.01611600	-1.67679100	-0.06993200
C	6.06085100	-3.60408700	-0.20473600
C	5.20651900	-1.42174500	-0.75575100
C	3.86851500	-2.93193600	0.52850200
C	4.87484400	-3.88057000	0.46252800
C	6.21674200	-2.36807400	-0.81588400
C	-0.53656500	0.00178400	0.00020600
C	-3.35236400	0.00309900	0.00050100
C	-1.25764500	-1.19810900	-0.00801400
C	-1.25650600	1.20232400	0.00855800
C	-2.63538600	1.20489900	0.00873300
C	-2.63649600	-1.19941000	-0.00789300
C	-4.77622000	0.00363200	0.00061200
C	-5.98843300	0.00345200	0.00060100
C	-8.43794500	1.06697900	-1.40561300
C	-8.44220600	0.68086900	1.62653200
C	-8.41927900	-1.75733000	-0.22318800
H	2.94195600	3.14512200	-1.04067600
H	5.33464300	0.46787200	1.25439700
H	7.13096600	2.13775200	1.35586200

H	4.73836700	4.84315700	-0.93883500
H	6.85364600	4.34209200	0.25461900
H	5.33427000	-0.46956300	-1.25417000
H	2.93895600	-3.14502600	1.04024800
H	4.73393300	-4.84461200	0.93843800
H	7.12921700	-2.14093800	-1.35547100
H	6.84985400	-4.34515700	-0.25453000
H	-0.70156000	-2.12633000	-0.01653900
H	-0.69956200	2.13002900	0.01696900
H	-3.17869200	2.14214800	0.01591700
H	-3.18070300	-2.13614300	-0.01500000
H	-9.53088000	1.08280800	-1.42460400
H	-8.08439300	0.68969400	-2.36714400
H	-8.08399400	2.09449600	-1.30017500
H	-9.53522100	0.68978200	1.64696000
H	-8.08987400	1.70255800	1.78207600
H	-8.08934700	0.07572700	2.46392800
H	-9.51181900	-1.79280400	-0.22707200
H	-8.05998500	-2.39796000	0.58461300
H	-8.06061700	-2.17546400	-1.16579600

4 (neutral, S_1)

Si	-7.81551500	0.00133900	-0.04809700
N	1.64444400	1.13965600	0.00954100
N	1.65404900	-1.09743200	0.20231200
C	0.86355500	0.04458500	0.09185100
C	2.96725300	-0.71039500	0.06482800
C	2.90583600	0.72486400	0.01831300
C	4.00457600	1.68728300	0.06568500
C	6.03247600	3.59486300	0.19540700
C	3.86767600	2.91180300	-0.59084700
C	5.16361000	1.44001600	0.80406200
C	6.16833100	2.39048900	0.86856800
C	4.87806100	3.85332400	-0.53259900
C	4.02012000	-1.65678100	-0.05400600
C	6.02962300	-3.59968200	-0.36289800
C	5.25062600	-1.34254700	-0.68009300
C	3.82859300	-2.98876400	0.39121000
C	4.82162600	-3.93273900	0.24084800
C	6.22987400	-2.29873900	-0.82239300
C	-0.53945100	0.02667700	0.10025500
C	-3.35918200	0.01537700	0.04812100
C	-1.27161000	-1.18193800	0.28002600
C	-1.27126000	1.23155200	-0.09110800
C	-2.63529500	1.22230400	-0.11699800
C	-2.63996900	-1.17909600	0.25200300
C	-4.76879600	0.00874400	0.01747400
C	-5.98359700	0.00487200	-0.00953300
C	-8.38804600	1.07394100	-1.46990800
C	-8.45756700	0.68683900	1.57054400
C	-8.40111600	-1.75971900	-0.28122700
H	1.33667200	-2.00781500	-0.09061500
H	2.95938800	3.10815300	-1.14477000
H	5.26691200	0.50941800	1.34629000
H	7.05793800	2.19099400	1.45210600
H	4.76585500	4.79503500	-1.05451000
H	6.82124100	4.33490000	0.24283400
H	5.40752100	-0.35232600	-1.08322800
H	2.91556900	-3.25706600	0.90865800
H	4.65970500	-4.93950300	0.60540500
H	7.15866300	-2.03716500	-1.31367700
H	6.80423600	-4.34593000	-0.48180200
H	-0.75424300	-2.11446200	0.46951400
H	-0.71973100	2.15335500	-0.21676900
H	-3.18397300	2.14308900	-0.26473300
H	-3.18894900	-2.10094500	0.39347200
H	-9.47963700	1.09962600	-1.51723100
H	-8.01540900	0.69307700	-2.42250500
H	-8.03009200	2.09872400	-1.35397400
H	-9.55051200	0.70109900	1.57529400
H	-8.10430100	1.70733700	1.72980800
H	-8.12195100	0.07952200	2.41313000
H	-9.49279200	-1.80259100	-0.30600400
H	-8.05596400	-2.39704000	0.53513700
H	-8.02406100	-2.17659000	-1.21696300

4 (anion, S_1)

Si	-7.77795000	-0.00135700	0.00079600
N	1.65270300	-1.14646100	0.01944300
N	1.65260400	1.14715800	-0.02077500
C	0.90618200	0.00031300	-0.00105400
C	2.91157500	-0.72515400	0.00178200
C	2.91152400	0.72598600	-0.00180900
C	4.01331700	1.68070300	0.06418300
C	6.05214400	3.60391200	0.21784500
C	3.84884400	2.95354800	-0.49888400
C	5.21301500	1.40718600	0.73096000
C	6.21892900	2.35448400	0.80211300
C	4.85746500	3.89688300	-0.42790600
C	4.01354200	-1.67978600	-0.06313300
C	6.05270300	-3.60275100	-0.21484800
C	5.21369900	-1.40623800	-0.72903600

C	3.84878600	-2.95251400	0.50006600
C	4.85757800	-3.89574700	0.43005100
C	6.21978900	-2.35342600	-0.79923800
C	-0.51908000	0.00030200	-0.00180800
C	-3.35570900	0.00053200	-0.00351500
C	-1.25272500	-1.21895500	0.03581600
C	-1.25254700	1.21975500	-0.04031600
C	-2.61748300	1.22194300	-0.04152100
C	-2.61769400	-1.22106400	0.03535700
C	-4.75098600	0.00046300	-0.00433700
C	-5.97795300	-0.00153000	-0.00613900
C	-8.45567600	1.04786200	-1.40666300
C	-8.46092800	0.70525600	1.60686300
C	-8.41444600	-1.75911400	-0.19117900
H	2.91027300	3.17805900	-0.98804300
H	5.34704500	0.44717000	1.21122500
H	7.13668400	2.11949600	1.32820600
H	4.71068300	4.87187200	-0.87792300
H	6.84019500	4.34496700	0.27572100
H	5.34796800	-0.44628200	-1.20935600
H	2.90986100	-3.17704400	0.98853700
H	4.71057500	-4.87066400	0.88014800
H	7.13792100	-2.11843000	-1.32466800
H	6.84089300	-4.34371700	-0.27196900
H	-0.69771000	-2.14806600	0.06033500
H	-0.69738500	2.14878500	-0.06419900
H	-3.16330200	2.15707400	-0.06994300
H	-3.16369500	-2.15610400	0.06303000
H	-9.54935200	1.07304000	-1.38922100
H	-8.13559100	0.65298200	-2.37289000
H	-8.09050000	2.07437900	-1.32929700
H	-9.55510700	0.72171000	1.59799600
H	-8.10582500	1.72702700	1.75804600
H	-8.13290700	0.10982300	2.46131400
H	-9.50769800	-1.78370400	-0.18092900
H	-8.05128200	-2.39111400	0.62194700
H	-8.07078400	-2.19422400	-1.13177500

5 (neutral, S_0)

N	0.37473700	1.10330400	-0.06028800
N	0.31367100	-1.08715300	-0.00331100
C	1.10532500	0.01615700	-0.02738300
C	-2.09515800	-1.64088700	0.09064800
C	-4.16859100	-3.50053800	0.29038000
C	-3.17945500	-1.40612700	0.93561400
C	-2.06513100	-2.82222900	-0.64974900
C	-3.09214200	-3.74710700	-0.54685100
C	-4.20900500	-2.32569800	1.02838200
C	-0.99614100	-0.67622300	0.00001800
C	-0.93273000	0.69914800	-0.05302800
C	-2.02372200	1.68182600	-0.12070400
C	-4.06372400	3.58596300	-0.22974800
C	-1.86548400	2.92668800	0.48491400
C	-3.21164300	1.41146100	-0.79711500
C	-4.22491900	2.35437700	-0.84665600
C	-2.87798300	3.86987700	0.43177100
C	2.56700900	-0.02141000	-0.01972400
C	5.36709900	-0.01914700	0.01377400
C	3.28085000	-1.20793100	-0.17473000
C	3.27335100	1.16953600	0.14501400
C	4.65260700	1.17199400	0.16106700
C	4.66251700	-1.21017500	-0.15606100
C	6.79613300	-0.01806400	0.03283100
C	7.99508600	-0.01651700	0.04901400
H	0.61187400	-2.03702800	0.12588900
H	-0.93598600	3.14588800	0.99388300
H	-3.33955300	0.46008600	-1.29727300
H	-5.14057500	2.12929300	-1.37907000
H	-2.74071300	4.83186300	0.90966200
H	-4.85511200	4.32366900	-0.27121800
H	-3.20843200	-0.49569400	1.51988100
H	-1.24374200	-3.00719600	-1.33235400
H	-3.05569900	-4.65731400	-1.13192300
H	-5.04447100	-2.12814200	1.68788400
H	-4.97369800	-4.21988600	0.36775100
H	2.76648000	-2.14865400	-0.32984100
H	2.71766600	2.09046500	0.25787400
H	5.19273300	2.10054700	0.28991500
H	5.20705500	-2.13683100	-0.27898700
H	9.05841900	-0.01395700	0.06305000

5 (anion, S_0)

N	0.34204900	1.13234000	0.00901000
N	0.34201200	-1.13203000	-0.00914000
C	1.06101200	0.00015600	-0.00005900
C	-0.93926200	-0.70368500	0.00123900
C	-0.93922600	0.70401700	-0.00149400
C	-2.03504200	1.67684100	-0.06946100
C	-4.07961500	3.60495700	-0.20413300
C	-1.88653500	2.93284500	0.52721300
C	-3.22640400	1.42172400	-0.75378700
C	-4.23642300	2.36827700	-0.81373000

C	-2.89258900	3.88172000	0.46131300
C	-2.03498100	-1.67672500	0.06926100
C	-4.07897200	-3.60546300	0.20443100
C	-3.22623300	-1.42200800	0.75391300
C	-1.88635400	-2.93261900	-0.52762300
C	-2.89210500	-3.88179100	-0.46148000
C	-4.23594900	-2.36886600	0.81415000
C	2.51948800	0.00014000	-0.00001900
C	5.33557300	0.00006400	0.00006400
C	3.23995000	-1.19931000	0.00841500
C	3.24004000	1.19957200	-0.00842200
C	4.61963600	1.20129600	-0.00839300
C	4.61957400	-1.20110200	0.00846900
C	6.76327500	-0.00001900	0.00010100
C	7.96570200	-0.00017500	0.00011500
H	-0.95616500	3.14617900	1.03741000
H	-3.35489600	0.46924300	-1.25141600
H	-5.14952400	2.14071600	-1.35212700
H	-2.75070300	4.84629700	0.93590900
H	-4.86841300	4.34626300	-0.25384700
H	-3.35490500	-0.46955200	1.25156000
H	-0.95608600	-3.14567600	-1.03812400
H	-2.75007100	-4.84625900	-0.93625000
H	-5.14891600	-2.14161100	1.35290200
H	-4.86750000	-4.34703600	0.25442800
H	2.68386200	-2.12757600	0.01690300
H	2.68398500	2.12785900	-0.01698700
H	5.16287500	2.13861100	-0.01564800
H	5.16275800	-2.13845000	0.01575800
H	9.02791500	-0.00024300	-0.00007000

5 (neutral, S_1)

N	0.32156400	1.13810600	0.03429400
N	0.31468000	-1.09505700	-0.21962700
C	1.09888100	0.04811300	-0.06524900
C	-1.00235800	-0.71238300	-0.07556300
C	-0.94187600	0.72823600	-0.00342900
C	-2.03490800	1.68913700	-0.06831100
C	-4.05954900	3.59932300	-0.23817900
C	-1.89110300	2.93314600	0.55337700
C	-3.20144800	1.42544500	-0.79276800
C	-4.20318500	2.37783000	-0.87693000
C	-2.89848100	3.87452800	0.47548100
C	-2.04865600	-1.65706400	0.04180300
C	-4.05459600	-3.60749300	0.35037200
C	-3.28494600	-1.34630500	0.66707800
C	-1.85145600	-2.99391000	-0.39863000
C	-2.84196900	-3.93846400	-0.24890700
C	2.50757500	0.03039800	-0.05829000
C	5.32175500	0.01477700	0.01947100
C	3.23699900	-1.16934500	-0.26887200
C	3.23485500	1.22451100	0.17924700
C	4.60135300	1.21340600	0.21698100
C	4.60858700	-1.16873200	-0.22812000
C	6.73893400	0.00729300	0.06314000
C	7.94132400	0.00374400	0.10106600
C	-4.26007300	-2.30592100	0.80764000
H	0.62823000	-2.00468900	0.08273300
H	-0.97937000	3.14138000	1.09705200
H	-3.31025300	0.48262900	-1.31189500
H	-5.09701100	2.16525500	-1.44930200
H	-2.78088100	4.82979900	0.97088500
H	-4.84645500	4.34020500	-0.30047600
H	-3.44421100	-0.35872600	1.07545500
H	-0.93747800	-3.26023500	-0.91544600
H	-2.67670300	-4.94576600	-0.61065400
H	-5.19038200	-2.04758200	1.29791900
H	-4.82699600	-4.35608500	0.46861100
H	2.72067200	-2.09449300	-0.49332500
H	2.68354900	2.14252900	0.33085500
H	5.14824400	2.12887500	0.40012600
H	5.15719700	-2.08675800	-0.39366500
H	9.00418700	0.00071000	0.13401100

5 (anion, S_1)

N	0.32790100	1.14814500	0.02018800
N	0.32789000	-1.14813000	-0.02020500
C	1.07626100	0.00001300	-0.00000100
C	-0.93073500	-0.72628900	-0.00225600
C	-0.93074300	0.72630900	0.00221800
C	-2.03241300	1.68026000	-0.06241900
C	-4.06920000	3.60743100	-0.21542000
C	-1.86374300	2.95600600	0.49518800
C	-3.23493300	1.40760100	-0.72547400
C	-4.23947500	2.35605700	-0.79594700
C	-2.87141000	3.90021300	0.42501700
C	-2.03240900	-1.68026300	0.06239000
C	-4.06916700	-3.60743300	0.21544700
C	-3.23482600	-1.40766900	0.72563800
C	-1.86381800	-2.95592500	-0.49540300
C	-2.87147700	-3.90014400	-0.42520100
C	-4.23936200	-2.35613200	0.79614500

C	2.49992600	-0.00000100	0.00001200	H	-4.85614500	4.34970800	-0.27319400
C	5.33660100	-0.00000500	0.00001600	H	-3.37154400	-0.44721300	1.20427600
C	3.23518300	-1.22050600	-0.04223700	H	-0.92277000	-3.18046900	-0.97975900
C	3.23518900	1.22050500	0.04226800	H	-2.72139200	-4.87645600	-0.87147600
C	4.60120000	1.21946100	0.04260000	H	-5.15889500	-2.12103900	1.31920200
C	4.60119300	-1.21947100	-0.04256600	H	-4.85610500	-4.34971600	0.27324500
C	6.74212600	-0.00000800	0.00000300	H	2.68142900	-2.15039300	-0.06911100
C	7.95536800	-0.00001500	-0.00000200	H	2.68143800	2.15039400	0.06914100
H	-0.92262800	3.18060600	0.97938700	H	5.14744700	2.15463800	0.07388200
H	-3.37172500	0.44710500	-1.20400800	H	5.14743500	-2.15465200	-0.07384600
H	-5.15907800	2.12090400	-1.31885300	H	9.01723600	-0.00006600	-0.00001900
H	-2.72125600	4.87657900	0.87115100				

Table S4. Effect of NaOH equivalents and reaction time on the conversion of **4** to **5**, as monitored by LCMS.[a]

NaOH (equiv)	Reaction monitoring time (min)	4 (%[b])	5 (%[b])
1	0	97.8	0.28
1	5	96.63	1.73
1	10	95.27	1.85
1	60	96.55	2.16
5	0	93.9	4.77
5	5	92.61	6.84
5	60	90.97	7.86
10	0	84.19	13.86
10	5	77.7	20.78
10	60	58.51	39.29
20	5	74.66	24.37
20	20	51.58	46.46
20	60	28.34	69.34

[a] Compound **4** (10 mg, 0.02547 mmol) was dissolved in DMSO (1 mL, 25.47 mM), followed by treatment with 1 N NaOH at 1, 5, 10, and 20 equiv. The reaction was monitored at the specified time intervals by LC–MS. [b] Values represent LCMS area percentages.

Table S5. Effect of HCl equivalents and reaction time on the conversion of **4** to **5**, as monitored by TLC and LCMS.[a]

HCl (equiv)	Reaction monitoring time (min)	4 (%[b])	5 (%[b])
1	5	97.70	0
1	30	97.99	0
10	5	No conversion[c]	-[c]
10	30	No conversion[c]	-[c]
20	5	No conversion[c]	-[c]
20	30	No conversion[c]	-[c]

[a] Compound **4** (10 mg, 0.02547 mmol) was dissolved in DMSO (1 mL, 25.47 mM), followed by treatment with 1 N HCl at 1, 10 and 20 equiv. The reaction was monitored at the specified time intervals by LCMS. [b] Values represent LCMS area percentages. [c] TLC used to confirm no conversion.